LIFTING A BOUNDED ABOVE ACYCLIC COMPLEX OF MODULES OF THE FORM P OVER IP TO AN ACYCLIC COMPLEX WHOSE TERMS LIE IN A CLASS STABLE UNDER SUMMANDS AND SURJECTIVITY MODULO THE IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 19342-19357 and the counted proof is the proof environment at source lines 19359-19387, which immediately follows it and is its own proof.

Editorial note. The proof displays its induction diagram with Xy-pic. The retained preamble carries the Xy-pic setup of the shared Stacks preamble, so the diagram is compiled from the author's own markup; nothing about the diagram was changed.

Editorial note. The chapter file pinned-original-source.tex carries only an input of the shared preamble on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus the native theorem declarations and mathematical macros, including Ker, Im, colim and the bold L of the derived tensor product). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

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Editorial note. The section lead-in sentence at source lines 19337-19340, which poses the general lifting question and announces that the acyclic case is discussed first, is left outside the counted spans because it is an announcement and not part of the lemma or its proof.

Allowed standard prerequisites (editorial).

- A bounded above acyclic complex of modules with a chosen isomorphism P^i equal to Z^i direct sum Z^{i+1} has differentials given by the projection Z^i direct sum Z^{i+1} to Z^{i+1} followed by the inclusion of Z^{i+1} into Z^{i+1} direct sum Z^{i+2}. recorded in the proof at source lines 19373-19376 as the form of the maps P^i -> P^{i+1}. Turns the acyclicity of the lifted complex and the vanishing of its cohomology into the bookkeeping of the modules Z^i, which is what property (2) of the statement is applied to.
- The lifting property of a projective module: a map into a quotient that factors through a surjection can be lifted, and the lift can be arranged to

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hit a prescribed element modulo the kernel. used at proof lines 19378-19382, where P^{n-1} is chosen projective and the composite P^{n-1} -> E^{n-1} -> E^n is lifted to a map P^{n-1} -> Z^n. Produces the new term P^{n-1} together with the differential into Z^n that extends the induction diagram one step to the left.

- The quotient map P -> P/IP is compatible with the given isomorphism P^i/IP^i -> E^i, so that the identification Z^n/IZ^n = Im(E^{n-1} -> E^n) holds. used at proof lines 19372 and 19380. Identifies the target of the lifted map with the quotient of Z^n, which is exactly the condition needed for the lifting property to apply.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Let $\mathcal{P}$ be a class of R-modules. Assume*

(1) *each $P \in \mathcal{P}$ is a projective R-module,*
(2) *if $P_1 \in \mathcal{P}$ and $P_1 \oplus P_2 \in \mathcal{P}$, then $P_2 \in \mathcal{P}$, and*
(3) *if $f : P_1 \to P_2$, $P_1, P_2 \in \mathcal{P}$ is surjective modulo I, then f is surjective.*

Then given any bounded above acyclic complex $E^\bullet$ whose terms are of the form P/IP for $P \in \mathcal{P}$ there exists a bounded above acyclic complex $P^\bullet$ whose terms are in $\mathcal{P}$ lifting $E^\bullet$.

Proof. Say $E^i = 0$ for $i > b$. Assume given n and a morphism of complexes

$$\begin{array}{ccccccccccc} & & & & P^n & \longrightarrow & P^{n+1} & \longrightarrow \ldots \longrightarrow & P^b & \longrightarrow & 0 \longrightarrow \ldots \\ & & & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \ldots & \longrightarrow & E^{n-1} & \longrightarrow & E^n & \longrightarrow & E^{n+1} & \longrightarrow \ldots \longrightarrow & E^b & \longrightarrow & 0 \longrightarrow \ldots \end{array}$$

with $P^i \in \mathcal{P}$, with $P^n \to P^{n+1} \to \ldots \to P^b$ acyclic in degrees $\geq n+1$, and with vertical maps inducing isomorphisms $P^i/IP^i \to E^i$. In this situation one can inductively choose isomorphisms $P^i = Z^i \oplus Z^{i+1}$ such that the maps $P^i \to P^{i+1}$ are given by $Z^i \oplus Z^{i+1} \to Z^{i+1} \to Z^{i+1} \oplus Z^{i+2}$. By property (2) and arguing inductively we see that $Z^i \in \mathcal{P}$. Choose $P^{n-1} \in \mathcal{P}$ and an isomorphism $P^{n-1}/IP^{n-1} \to E^{n-1}$. Since P^{n-1} is projective and since $Z^n/IZ^n = \mathrm{Im}(E^{n-1} \to E^n)$, we can lift the map $P^{n-1} \to E^{n-1} \to E^n$ to a map $P^{n-1} \to Z^n$. By property (3) the map $P^{n-1} \to Z^n$ is surjective. Thus we obtain an extension of the diagram by adding P^{n-1} and the maps just constructed to the left of P^n. Since a diagram of the desired form exists for $n > b$ we conclude by induction on n. □